

APPM 5720

Open Topics in Applied Mathematics

FORMAT	CATEGORY	CREDITS
On-Campus	Elective	None
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective (On-Campus)	Data Scientist (33.4 / 100)	27.7 / 100

28	66	+51
CURRENT AVG SCORE across 6 job roles	POTENTIAL SCORE with recommended additions	MAX SCORE GAIN best single-role improvement

CURRENT COURSE DESCRIPTION

Provides a vehicle for the development and presentation of new topics that may be incorporated into the core courses in applied mathematics.
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INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
Data Scientist	33.4	80.5	+47.1	HIGH
ML Engineer	30.1	53.4	+23.3	MEDIUM
Applied Scientist	29.8	80.7	+50.9	HIGH
AI Engineer	29.4	70.5	+41.1	HIGH
Machine Learning Engineer	29.1	66.7	+37.6	HIGH
MLOps Engineer	14.3	46.9	+32.6	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (example) appear as gaps across 3 or more job roles.

System Design

4 of 6 roles

systems

Business Communication

1 of 6 roles

insights

Deep Learning

1 of 6 roles

training

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

1 **HIGH** **Add System Design coverage** — required vocabulary in 4 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.

2 **LOW** **Optional: Business Communication** — would broaden appeal to one additional role. Candidate terms: insights.

3 **LOW** **Optional: Deep Learning** — would broaden appeal to one additional role. Candidate terms: training.

GAP KEYWORDS BY ROLE

Data Scientist

33.4 → 80.5 (+47.1 pts)

data sets insights different

ML Engineer

30.1 → 53.4 (+23.3 pts)

data learning machine systems models world building detection

Applied Scientist

29.8 → 80.7 (+50.9 pts)

biology computational research literature models agents department drug

AI Engineer

29.4 → 70.5 (+41.1 pts)

following data robotics models learning software systems based

Machine Learning Engineer

29.1 → 66.7 (+37.6 pts)

data sets learning design science models machine systems

MLOps Engineer

14.3 → 46.9 (+32.6 pts)

detection data systems learning machine image training models

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.

